



BY

P.Hamsika 2520030586

M.Thanishka 2520030583

DBSE&DBD Project

Women Safety

SOS & Location Tracking

UNDER THE GUIDENCE
OF

DR.RAMYA KRISHNA

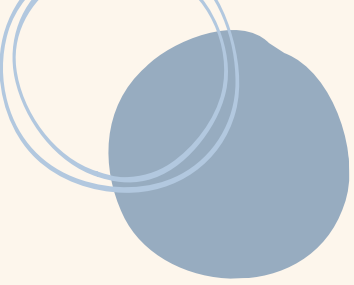
ABSTRACT

- The Women Safety Application is a web-based platform designed to provide multiple personal safety and emergency-support features in one system.
- It provides Normal SOS, Secret SOS, Emergency Contacts, Location Tracking, Safety Timer, Safety Check-in, Safe Zones, and Fake Call functionality.
- The application follows a full-stack architecture using React + Vite, Node.js + Express.js, and MySQL.
- The backend manages application logic, authentication, API communication, and database operations.
- The application is deployed on Railway, making it accessible through a web browser.

PROBLEM STATEMENT

- Emergency situations may require users to communicate quickly with trusted people.
- Manually finding contacts, communicating the situation, and sharing location can require multiple steps.
- Using separate applications for emergency communication, location sharing, and safety monitoring can increase complexity.
- A centralized platform is needed to provide multiple safety-support mechanisms through a single accessible application.

EXISTING VS PROPOSED SYSTEM

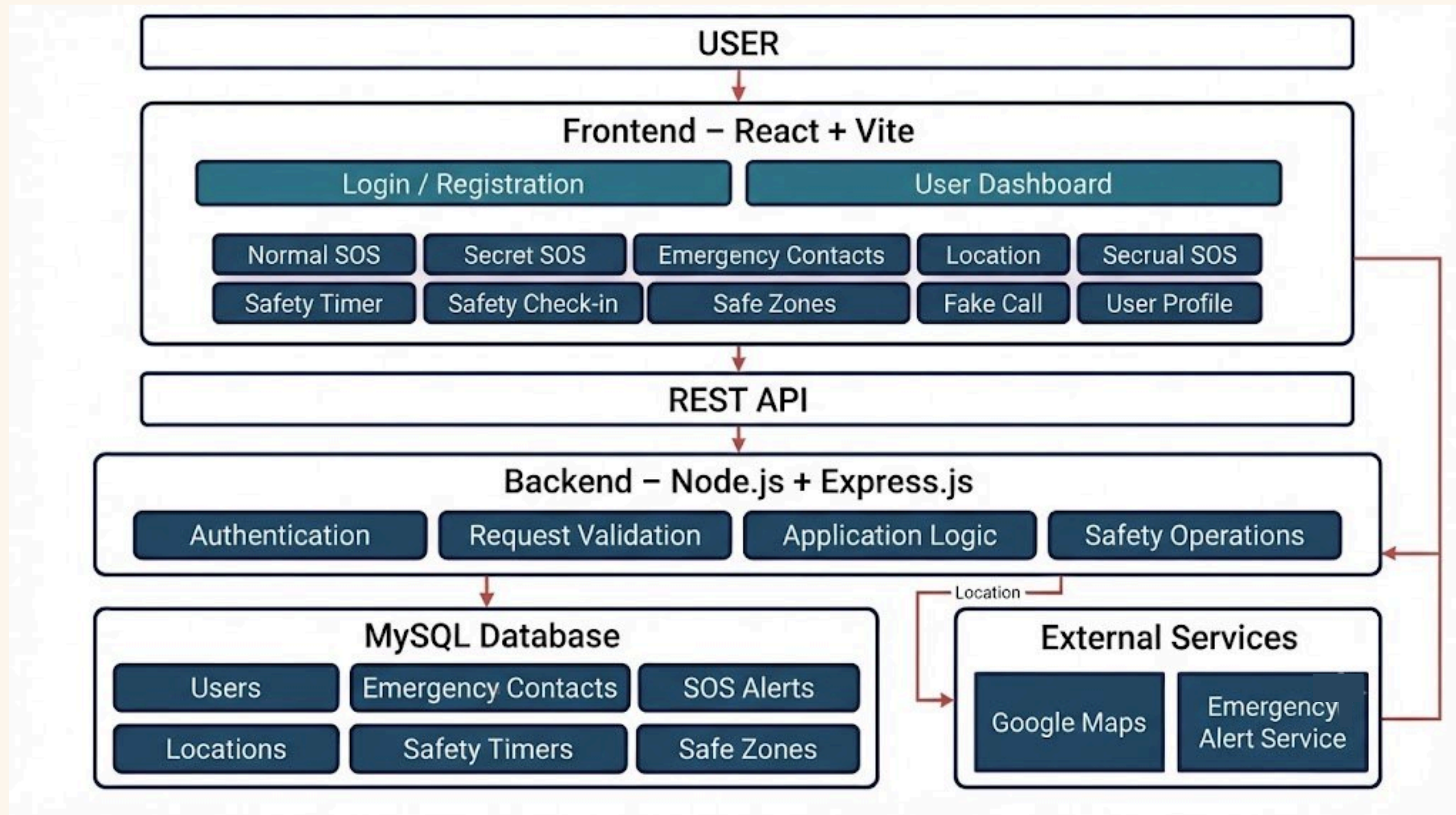


Existing Approach	Proposed System
Safety-related actions may involve multiple applications or manual steps.	Multiple safety-support functions are available in one web application.
Emergency contacts may need to be accessed separately.	Users can manage trusted emergency contacts within the application.
Emergency alert and location functions may be handled separately.	SOS features are integrated with location-related functionality.
Safety monitoring may require separate tools.	Safety Timer and Safety Check-in are included in the platform.
Users may need different tools for different situations.	Normal SOS, Secret SOS, Safe Zones and Fake Call provide different safety mechanisms.
Data and application operations may be distributed across separate systems.	React frontend, Express backend and MySQL database work as an integrated system.

Objectives

- Develop an accessible and user-friendly Women Safety Application.
- Provide secure authentication and user account management.
- Manage trusted emergency contacts and SOS mechanisms.
- Provide location-based safety functionality and monitoring.
- Integrate Safety Timer, Check-in, Safe Zones and Fake Call.
- Build and deploy a complete full-stack system using frontend, backend, database and APIs.

System Architecture



Technology Stack

Layer	Technology	Purpose
Frontend	React	User interface
Build Tool	Vite	Frontend development and
Programming	JavaScript	Application logic
Backend	Node.js + Express.js	Server and REST APIs
Database	MySQL	Persistent data
API Testing	Postman	Backend API testing
Version Control	Git + GitHub	Source-code management
Development	Visual Studio Code	Application
Deployment	Railway	Online hosting
Location	Browser/Device Location Services	Location-related functionality

Module Identification

User Management

- Registration
- Login & Authentication
- Profile Management

Emergency Management

- Emergency Contacts
- Normal SOS
- Secret SOS

Safety Tools

- Fake Call

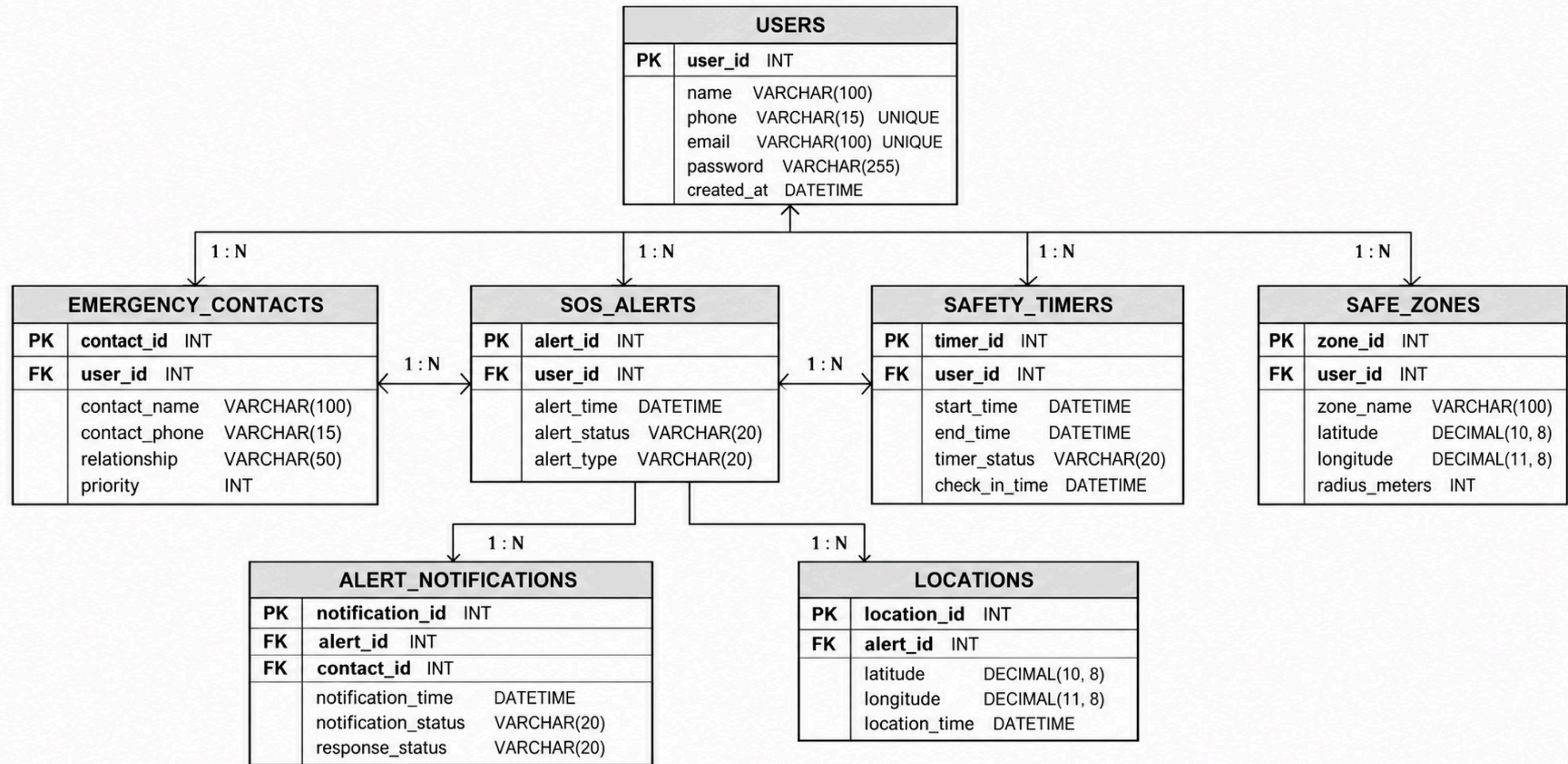
Safety & Location

- Location / Live Tracking
- Safety Timer
- Safety Check-in
- Safe Zones

Backend & Data Management

- REST API
- Authentication & Authorization
- MySQL Database
- API Integration

ER Diagram



Implementation & Key Features

Implementation

- React + Vite frontend
- Node.js + Express.js backend
- MySQL database
- REST APIs
- JWT authentication

Key Features

- Normal & Secret SOS
- Emergency Contacts
- Location
- Safety Timer & Check-in
- Safe Zones
- Fake Call

Testing & Deployment

Testing

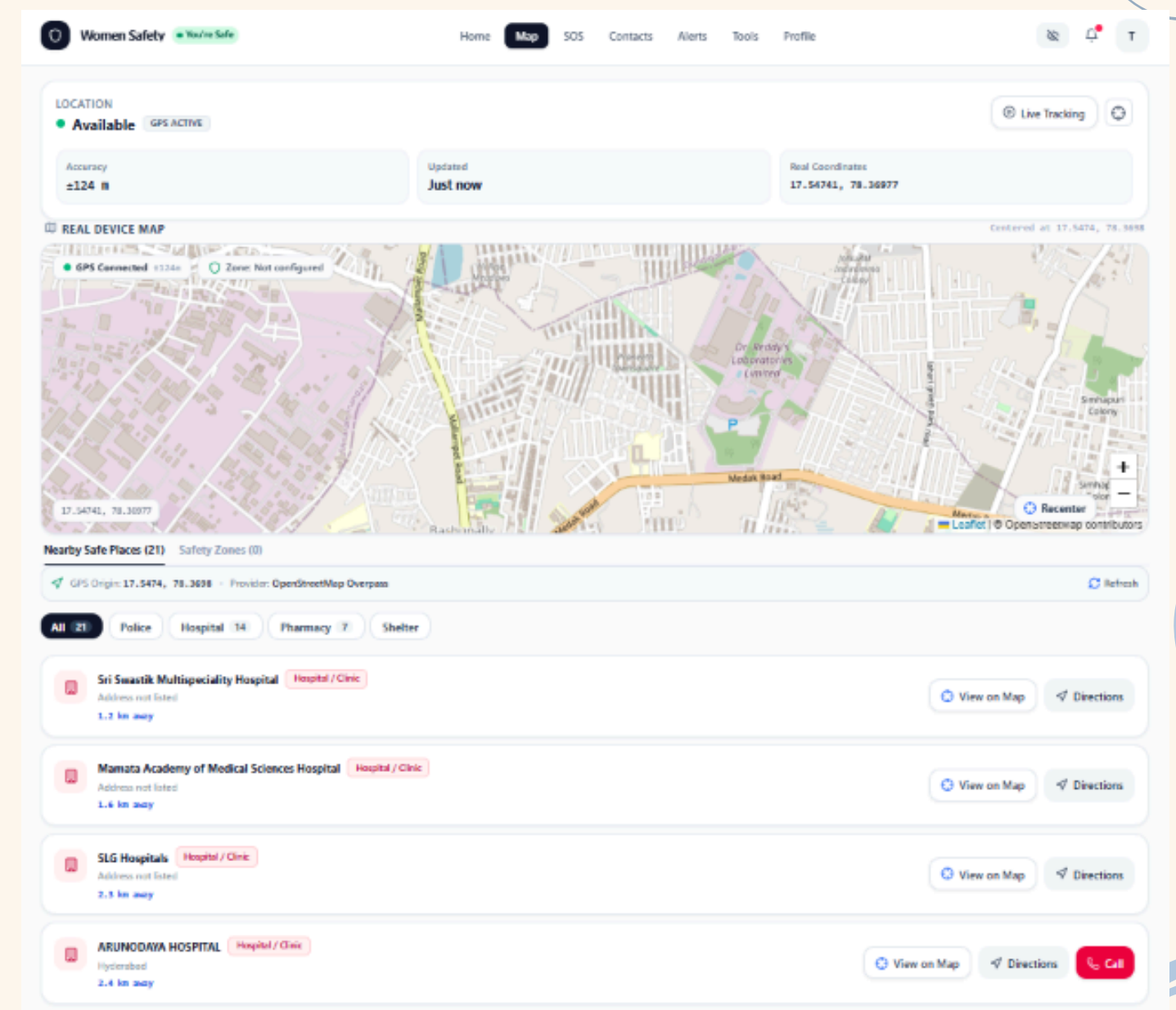
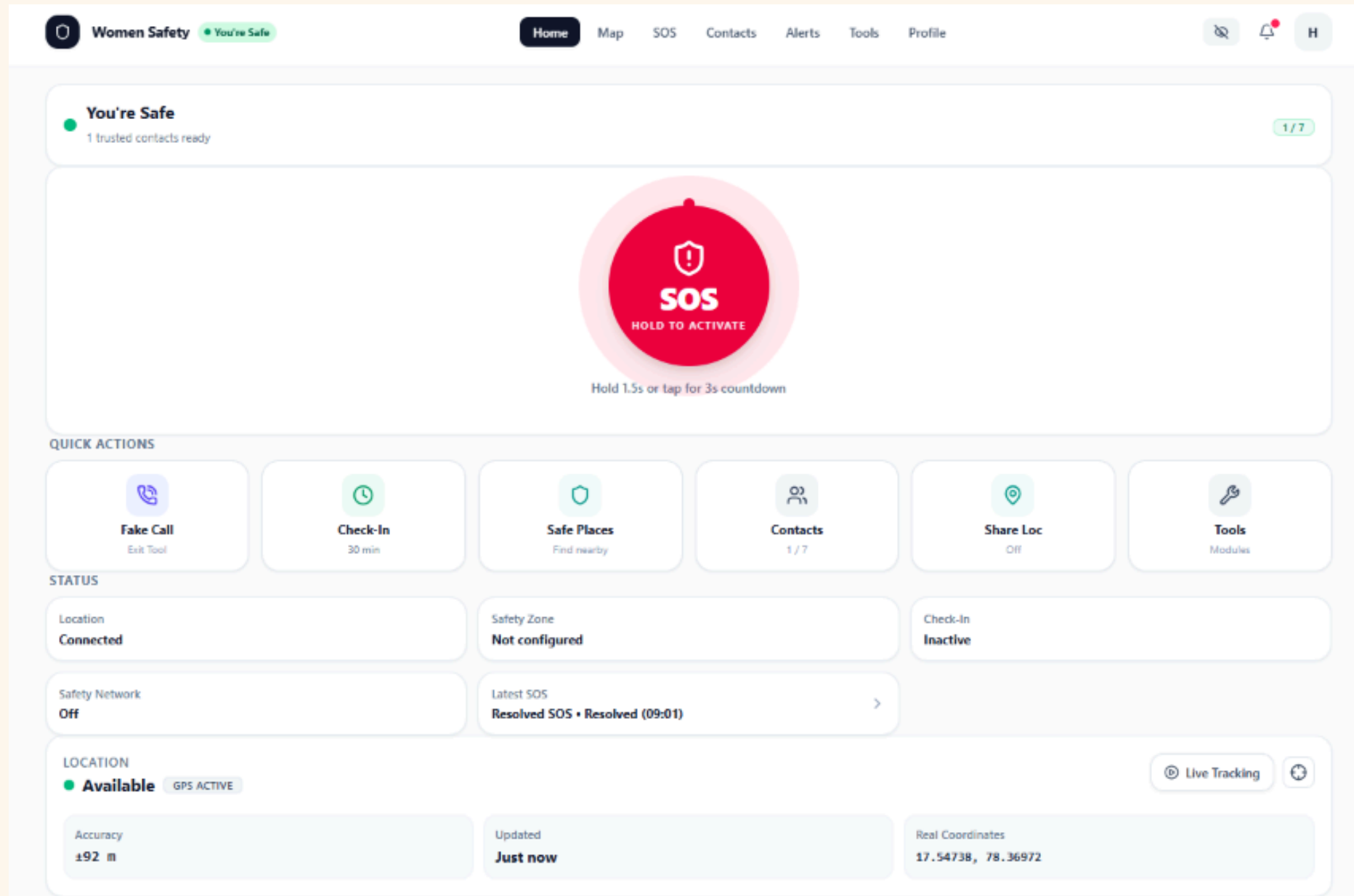
- Postman API testing
- Browser testing
- MySQL verification
- 15 documented test cases

Deployment

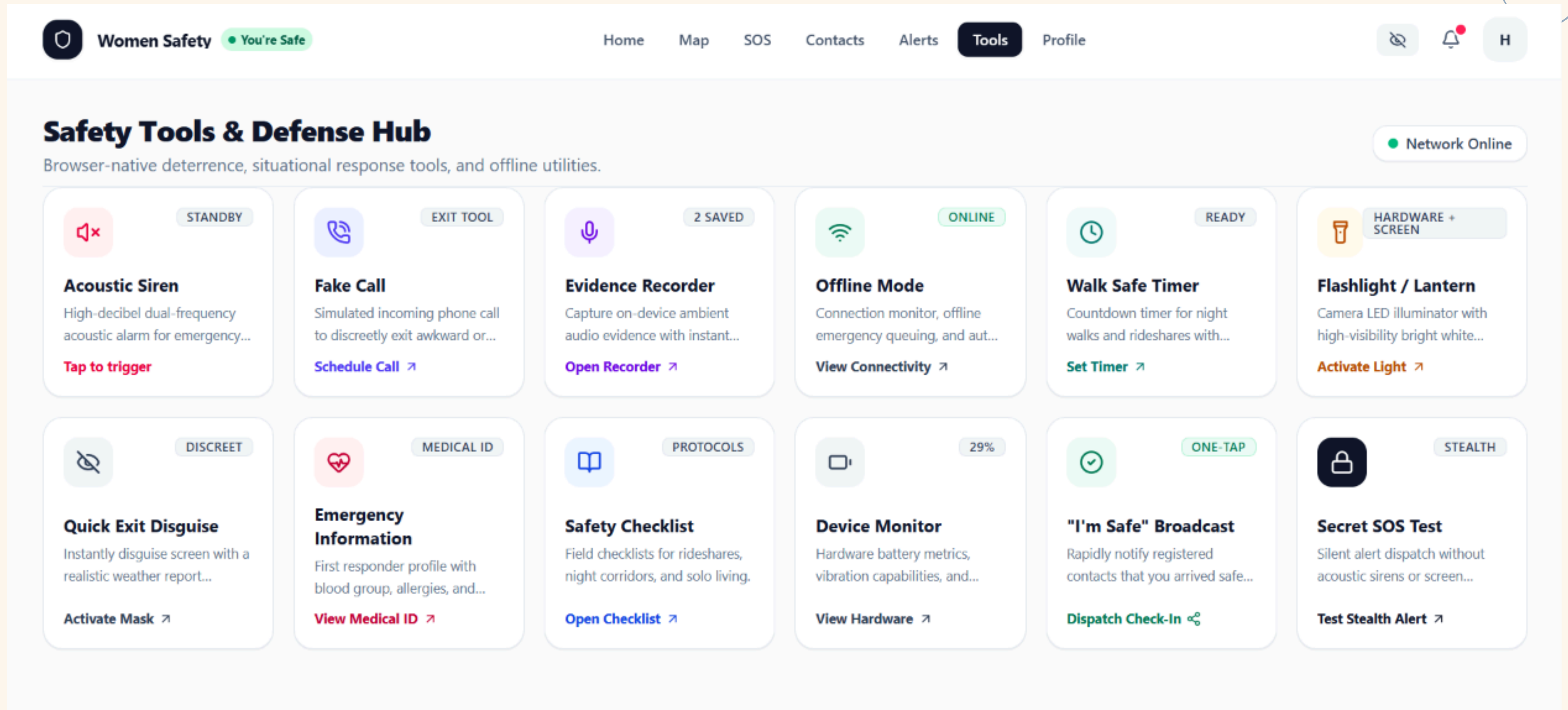
- GitHub → Railway
- Production environment configured
- Live application tested

Live Prototype: dbse-and-dbd-project-production.up.railway.app:

Prototype



Prototype



Future Scope & Conclusion

Future Scope

- Android/iOS application
- SMS/push notifications
- Improved offline support
- Enhanced location and emergency integration

Conclusion

- Developed a full-stack Women Safety application with integrated emergency and safety features.
- Implemented authentication, emergency contacts, SOS, location, safety monitoring, safe zones and fake call functionality.
- Integrated React, Node.js, Express.js and MySQL into a working system.
- Tested and deployed the application for practical use and future enhancement.

The background features a light beige color with abstract geometric patterns in shades of blue. These patterns include solid circles of various sizes, some filled with small dots, and others with concentric rings or thin outlines. The shapes are scattered across the corners and edges of the frame, creating a modern, minimalist aesthetic.

Thank you